PEMROGRAMAN BERBASIS FRAMEWORK
JOBSHEET 8

NAMA : Muhammad Fahreza Rohmansyah
KELAS : 3E
NO. ABSEN : 15
NIM : 2341720151

PROGRAM STUDI TEKNIK INFORMATIKA
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI MALANG
2026

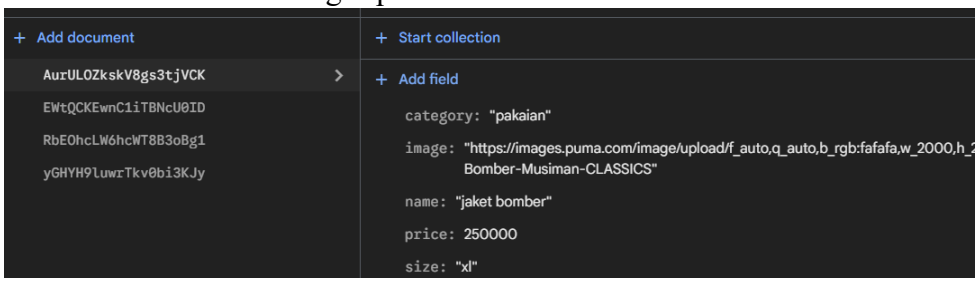
Link Github:

https://github.com/rezfahreza/PBF_2026.git

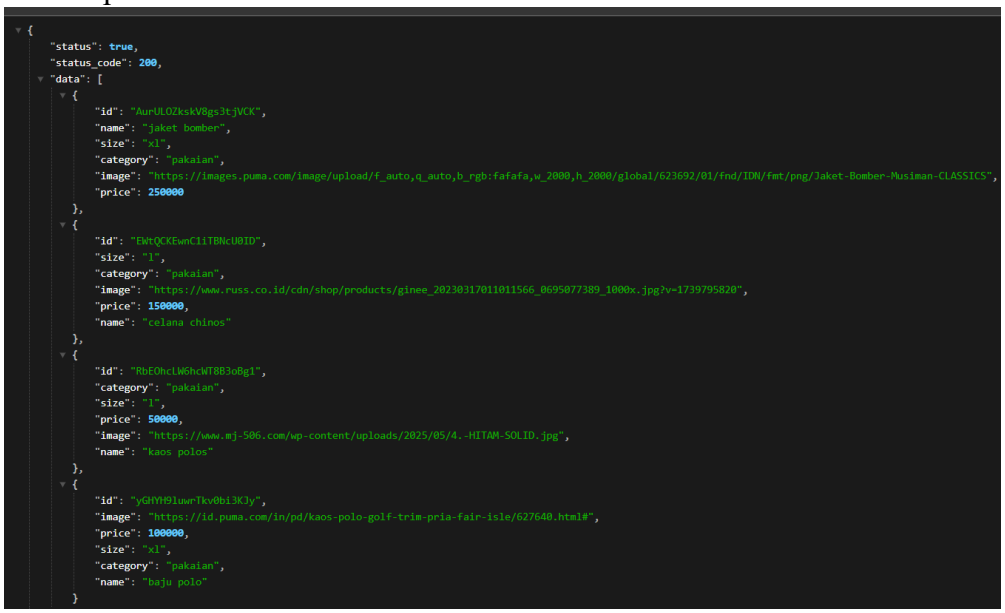
Praktikum

1. Setup Data Produk

- Tambahkan field images pada firebase



- Cek pada browser



2. Implementasi CSR dengan useEffect

- Membuat file index.tsx pada folder views/products

JS_8 > my-app > src > pages > view > produk > index.tsx > default

```tsx
type ProductType = {
    id: string;
    image: string;
    name: string;
    price: number;
    size: string;
    category: string;
};

const TampilanProduk = ({products}: {products: ProductType[]}) => {
    return (
        <div>
            <h1>Daftar Produk</h1>
            {products.map((product: ProductType) => (
                <div key={product.id}>
                    <h2>Nama: {product.name}</h2>
                    <p>Harga: Rp{product.price.toFixed(2)}</p>
                    <p>Ukuran: {product.size}</p>
                    <img src={product.image} alt={product.name} className="h-32 w-32 object-cover"/>
                    <p>Kategori: {product.category}</p>
                </div>
            ))}

        </div>
    );
};

export default TampilanProduk;
```

- Jalankan browser

- Buat file produk.modules.scss

JS_8 > my-app > src > pages > produk > produk.module.scss >

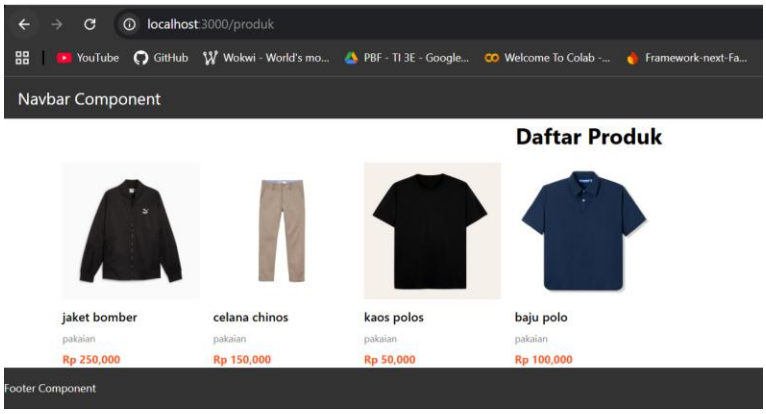
```scss
.produk {
  width: 100%;
  padding: 0 5%;
  &__title {
    text-align: center;
    font-size: 32px;
    font-weight: bold;
    margin-bottom: 16px;
  }
  &__content {
    display: flex;
    flex-wrap: wrap;
    gap: 20px;
    &__item {
      &__image {
        width: 100%;
        height: 200px;
        object-fit: cover;
        border-radius: 4px;
        margin-bottom: 12px;
      }
      &__name {
        font-size: 18px;
        font-weight: 600;
        margin-bottom: 8px;
      }
      &__price {
        font-size: 16px;
        color: #ff5722;
        font-weight: bold;
      }
      &__category {
        font-size: 14px;
        color: #888;
        margin-bottom: 8px;
      }
    }
  }
}
```

- Modifikasi Pada file index.tsx pada folder pages/views/product

JS_8 > my-app > src > pages > view > produk > index.tsx > ...

```tsx
import styles from "../../produk/product.module.scss";

type ProductType = {
    id: string;
    image: string;
    name: string;
    price: number;
    size: string;
    category: string;
};

const TampilanProduk = ({products}: {products: ProductType[]}) => {
    return (
        <div className={styles.produk}>
            <h1 className={styles.produk__title}>Daftar Produk</h1>
            <div className={styles.produk__content}>
                {products.map((products: ProductType) => (
                    <div key={products.id} className={styles.produk__content__item}>
                        <div className={styles.produk__content__item__image}>
                            <img src={products.image} alt={products.name} width={200} />
                        </div>
                        <h4 className={styles.produk__content__item__name}>
                            {products.name}
                        </h4>
                        <p className={styles.produk__content__item__category}>
                            {products.category}
                        </p>
                        <p className={styles.produk__content__item__price}>
                            Rp {products.price.toLocaleString()}
                        </p>
                    </div>
                ))}
            </div>
        </div>
    );
};

export default TampilanProduk;
```

- Jalankan browser



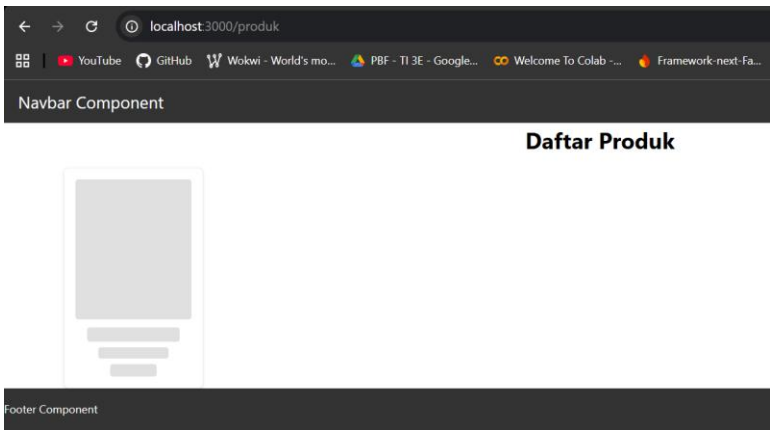
3. Implementasi Skeleton Loading

- Modfikasi file index.tsx pada folder views/product/index.tsx dan product.module.scss

```
const TampilanProduk = ({products}: {products: ProductType[]}) => {
    return (
        <div className={styles.produk}>
            <h1 className={styles.produk__title}>Daftar Produk</h1>
            <div className={styles.produk__content}>
                {/*{products.map((products: ProductType) => (
                    <div key={products.id} className={styles.produk__content__item}>
                        <div className={styles.produk__content__item__image}>
                            <img src={products.image} alt={products.name} width={200} />
                        </div>
                        <h4 className={styles.produk__content__item__name}>
                            {products.name}
                        </h4>
                        <p className={styles.produk__content__item__category}>
                            {products.category}
                        </p>
                        <p className={styles.produk__content__item__price}>
                            Rp {products.price.toLocaleString()}
                        </p>
                    </div>
                ))}*/}
                <div className={styles.produk__content__skeleton}>
                    <div className={styles.produk__content__skeleton__image}> </div>
                    <div className={styles.produk__content__skeleton__name}></div>
                    <div className={styles.produk__content__skeleton__category}></div>
                    <div className={styles.produk__content__skeleton__price}></div>
                </div>
            </div>
        </div>
    );
};
```

```
JS_8 > my-app > src > pages > produk > product.module.scss > .produk >
.produk {
  &__content {
    flex-wrap: wrap;
    gap: 20px;
    &__skeleton {
      width: 200px;
      padding: 16px;
      border: 1px solid #eee;
      border-radius: 8px;
      box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
      display: flex;
      flex-direction: column;
      align-items: center;
      animation: identifier 1.5s infinite ease-in-out;
      &__image {
        width: 100%;
        height: 200px;
        background-color: #e0e0e0;
        border-radius: 4px;
        margin-bottom: 12px;
      }
      &__name {
        width: 80%;
        height: 20px;
        background-color: #e0e0e0;
        border-radius: 4px;
        margin-bottom: 8px;
      }
      &__category {
        width: 60%;
        height: 16px;
        background-color: #e0e0e0;
        border-radius: 4px;
        margin-bottom: 8px;
      }
      &__price {
        width: 40%;
        height: 18px;
        background-color: #e0e0e0;
        border-radius: 4px;
      }
    }
    &__item {
      &__image {
        width: 100%;
        height: 200px;
        object-fit: cover;
        border-radius: 4px;
        margin-bottom: 12px;
      }
```

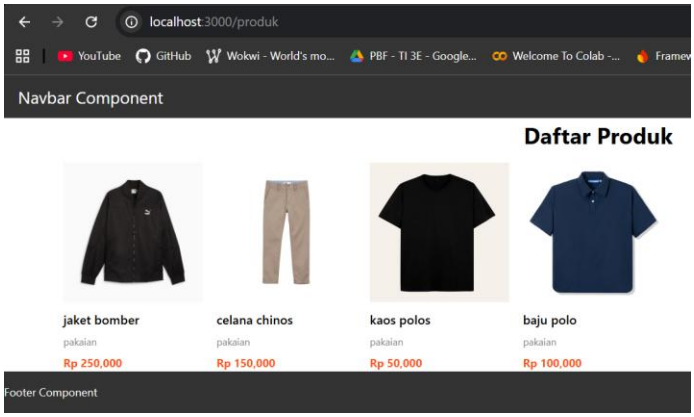
- Jalankan browser



- Modifikasi pada index.tsx pada folder views/product/index.tsx

```
JS_8 > my-app > src > pages > view > produk > index.tsx > TampilanProduk
const TampilanProduk = ({products}: {products: ProductType[]}) => {
        <div className={styles.produk}>
            <h1 className={styles.produk__title}>Daftar Produk</h1>
            <div className={styles.produk__content}>
                {products.length > 0 ? (
                <>
                    {products.map((products: ProductType) => (
                        <div key={products.id} className={styles.produk__content__item}>
                            <div className={styles.produk__content__item__image}>
                                <img src={products.image} alt={products.name} width={200} />
                            </div>
                            <h4 className={styles.produk__content__item__name}>
                                {products.name}
                            </h4>
                            <p className={styles.produk__content__item__category}>
                                {products.category}
                            </p>
                            <p className={styles.produk__content__item__price}>
                                Rp {products.price.toLocaleString()}
                            </p>
                        </div>
                    ))}
                </>
                ) : (
                <div className={styles.produk__content__skeleton}>
                    <div className={styles.produk__content__skeleton__image}></div>
                    <div className={styles.produk__content__skeleton__name}></div>
                    <div className={styles.produk__content__skeleton__category}></div>
                    <div className={styles.produk__content__skeleton__price}></div>
                </div>
                )}
            </div>
        </div>
    );
};
```

- Jalankan browser



4. Implementasi SWR

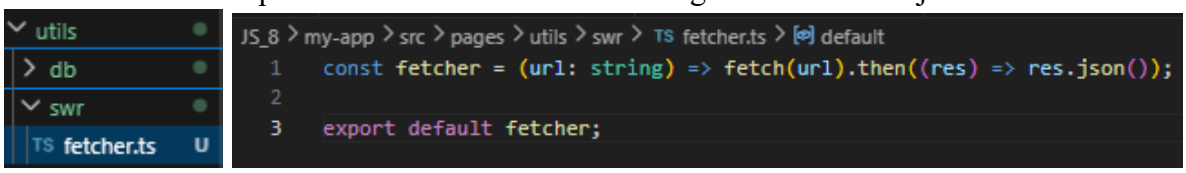
- Install SWR

```
PS E:\PBF_2026\JS_8\my-app> npm install swr

added 3 packages, and audited 471 packages in 31s

154 packages are looking for funding
  run `npm fund` for details
```

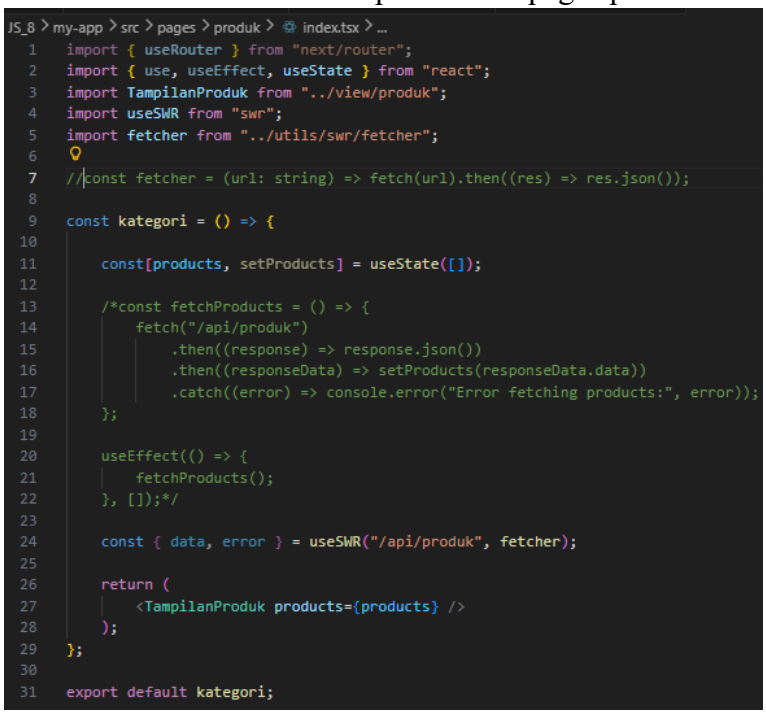
- Buka dan modifkasi file index.tsx pada folder pages/product/

```
JS_8 > my-app > src > pages > produk > index.tsx > kategori
import { useRouter } from "next/router";
import { use, useEffect, useState } from "react";
import TampilanProduk from "../view/produk";
import useSWR from "swr";

const fetcher = (url: string) => fetch(url).then((res) => res.json());

const kategori = () => {

    const[products, setProducts] = useState([]);

    /*const fetchProducts = () => {
        fetch("/api/produk")
            .then((response) => response.json())
            .then((responseData) => setProducts(responseData.data))
            .catch((error) => console.error("Error fetching products:", error));
    };

    useEffect(() => {
        fetchProducts();
    }, []);*/

    const { data, error } = useSWR("/api/produk", fetcher);

    return (
        <TampilanProduk products={products} />
    );
};

export default kategori;
```

- Buat folder swr pada utils dan tambahkan file dengan nama fetcher.js



- Modifikasi file index.tsx pada folder pages/produk



Tugas Praktikum

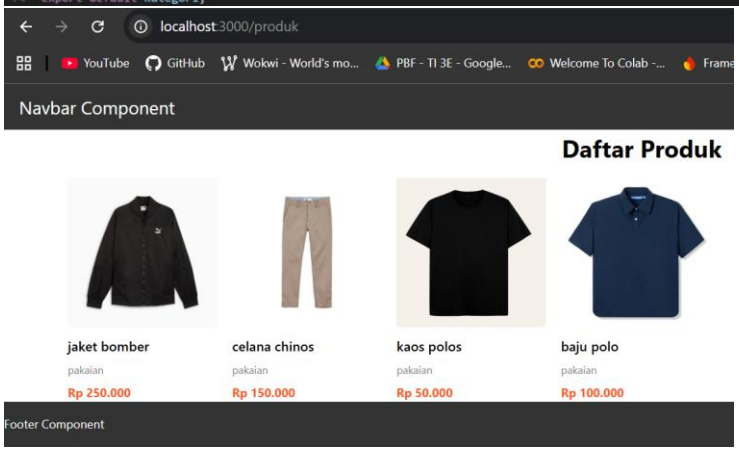
1. Jelaskan perbedaan:

- Client Side Rendering :
  Proses perenderan UI dilakukan sepenuhnya di browser (client). Server mengirimkan HTML kosong dan JavaScript, lalu browser mengambil data dari API untuk menampilkan konten. Ada jeda (*loading*) sebelum data muncul
- Server Side Rendering :
  Proses perenderan HTML dilakukan di server setiap kali ada permintaan dari user. Browser menerima HTML yang sudah berisi konten lengkap, sehingga lebih cepat terlihat oleh pengguna dan lebih baik untuk SEO
- Static Site Generation :
  Halaman web dibuat (di-*generate*) menjadi file statis pada saat waktu build (build time). Halaman yang sudah jadi ini disimpan dan dikirim langsung ke user tanpa perlu diproses ulang di server, menjadikannya metode yang paling cepat

2. Buat halaman produk dengan:

- Skeleton loading
- Animasi

```python
import { useEffect, useState } from "react";
import styles from "./product.module.scss";

type ProductType = {
  id: string;
  name: string;
  price: number;
  image: string;
  size: string;
  category: string;
};

const Kategori = () => {
  const [products, setProducts] = useState<ProductType[]>([]);
  const [loading, setLoading] = useState(true);

  const fetchProducts = () => {
    setLoading(true);

    fetch("/api/produk")
      .then((res) => res.json())
      .then((resData) => {
        setProducts(resData.data);
        setLoading(false);
      })
      .catch((err) => {
        console.error(err);
        setLoading(false);
      });
  };

  useEffect(() => {
    fetchProducts();
  }, []);

  return (
    <div className={styles.produk}>
      <h1 className={styles.produk__title}>Daftar Produk</h1>

      <div className={styles.produk__content}>
        {loading
          ? Array.from({ length: 4 }).map((_, index) => (
              <div key={index} className={styles.produk__content__skeleton}>
                <div className={styles.produk__content__skeleton__image + " " + styles["animate-pulse"]}></div>
                <div className={styles.produk__content__skeleton__name + " " + styles["animate-pulse"]}></div>
                <div className={styles.produk__content__skeleton__category + " " + styles["animate-pulse"]}></div>
                <div className={styles.produk__content__skeleton__price + " " + styles["animate-pulse"]}></div>
              </div>
            ))
          : products.map((product) => (
              // Menambahkan inline width agar lebarnya sama dengan skeleton
              <div key={product.id} className={styles.produk__content__item} style={{ width: "200px" }}>
                <img
                  src={product.image}
                  alt={product.name}
                  className={styles.produk__content__item__image}
                />
                <h2 className={styles.produk__content__item__name}>
                  {product.name}
                </h2>
                <p className={styles.produk__content__item__category}>
                  {product.category}
                </p>
                <p className={styles.produk__content__item__price}>
                  Rp {product.price.toLocaleString("id-ID")}
                </p>
              </div>
            ))}
      </div>
    </div>
  );
};

export default Kategori;
```



3. Refactor kode dari useEffect menjadi SWR

```
import useSWR from "swr";
import styles from "./product.module.scss";
import fetcher from "../utils/swr/fetcher";

type ProductType = {
  id: string;
  name: string;
  price: number;
  image: string;
  size: string;
  category: string;
};

const Kategori = () => {
  const { data, error, isLoading } = useSWR("/api/produk", fetcher);

  const products: ProductType[] = data?.data || [];

  if (error) {
    return <p>Gagal mengambil data</p>;
  }

  return (
    <div className={styles.produk}>
      <h1 className={styles.produk__title}>Daftar Produk</h1>

      <div className={styles.produk__content}>
        {isLoading
          ? Array.from({ length: 4 }).map((_, index) => (
              <div key={index} className={styles.produk__content__skeleton}>
                <div className={styles.produk__content__skeleton__image + " " + styles["animate-pulse"]}></div>
                <div className={styles.produk__content__skeleton__name + " " + styles["animate-pulse"]}></div>
                <div className={styles.produk__content__skeleton__category + " " + styles["animate-pulse"]}></div>
                <div className={styles.produk__content__skeleton__price + " " + styles["animate-pulse"]}></div>
              </div>
            ))
          : products.map((product) => (
              <div
                key={product.id}
                className={styles.produk__content__item}
                style={{ width: "200px" }}
              >
                <img
                  src={product.image}
                  alt={product.name}
                  className={styles.produk__content__item__image}
                />
                <h2 className={styles.produk__content__item__name}>
                  {product.name}
                </h2>
                <p className={styles.produk__content__item__category}>
                  {product.category}
                </p>
                <p className={styles.produk__content__item__price}>
                  Rp {product.price.toLocaleString("id-ID")}
                </p>
              </div>
            ))}
      </div>
    </div>
  );
};

export default Kategori;
```